Multilayer Network Analysis of Startup Cofounders and Investment Managers

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***Abstract*—Startup ecosystems are multilayer social networks where cofounders and investment managers connect through shared portfolio ties, education, work experience, and volunteering. This paper analyzes the portfolio of 212 (54 companies) as a multilayer network to identify community structure and the institutional signals that explain it. A bipartite cofounder-investment manager network is built and three institutional layers (education, experience, volunteering) are attached using LinkedIn-derived data. The final graph includes 59 nodes (49 for cofounders and 10 for investment managers) and 93 multilayer edges. Using Louvain, greedy modularity, label propagation, and Girvan-Newman algorithms, stable communities appear in the 5-to-8 range with modularity up to about 0.49. Network statistics show a sparse but connected structure, with density 0.054, average degree 3.15, diameter 6, and negative assortativity ($r = -0.718$); this disassortativity is partly structurally induced by the bipartite investment-manager–founder configuration and should be interpreted accordingly rather than as a purely emergent finding. Centrality rankings highlight investment managers as bridges across founder groups, while the cofounder side splits into sub-communities tied to shared institutional histories. Educational overlap concentrates around local universities with a smaller international tier, suggesting institutional pathways for access and trust formation. Implications for founders, investors, and policy are outlined, along with a reproducible workflow for multilayer network construction, community detection, and comparative analysis across layers.**

***Index Terms*—startup ecosystems, multilayer networks, venture capital, community detection, cofounder networks, investment managers, LinkedIn data**

I. Introduction

The venture capital industry operates on the basis of asymmetric information: success, in general, depends on privileged access to deal flow and the capability to identify founders with high potential before the competitors. Such advantages have conventionally been attributed to individual relationships or intuitions of the venture capitalists rather than a logical framework that considers the connections allowing the capital and the consequent opportunities to circulate. However, as startup ecosystems grow in scale and become more complex while maturing, a more rigorous account, grounded in the social and professional structures governing the aforementioned circulation of the capital and opportunities, is needed.

This study applies network science techniques in order to approach and analyze the portfolio of 212 [1], a prominent venture capital firm in Turkey, targeting tech-oriented startups in Turkey and other locations, primarily as a multilayer network in which various kinds of possible relationships, namely education, professional experience, and volunteering, between startup cofounders and investment managers are attempted to be captured through the conceptualization that each person, whether a cofounder or an investment manager, would be represented with a corresponding node, and each commonality between any two of the people would be represented with an edge connecting the nodes representing those two people. By mapping the connections between cofounders and investment managers via shared ties in the described categories by the means of common institutions, the study aims to uncover the latent community structures which may facilitate trust and collaboration in the startup ecosystem. Furthermore, a multilayer approach towards the portfolio allows observing the informal pathways that precede and sustain formal business relationships.

The main contributions of this study are a reproducible workflow to construct the dataset for the portfolio of 212 through the means of webscraping the website of the firm [2] and the LinkedIn pages whose links were obtained from the scraped data and to regenerate the dataset, networks, and the analysis results, the created dataset and the networks generated from that particular dataset, and the application of certain community detection algorithms on the multilayer network of cofounders and investment managers, which can be considered the *primary network outcome* of the study. The findings of the study reveal a highly disassortative organization ($r = -0.718$) with a core-periphery structure where investment managers act as bridges across groups of founders which would otherwise be isolated from each other—a pattern that is partly a structural consequence of the bipartite network design and should be interpreted as such alongside the community-level evidence. Ultimately, the study aims to demystify the circulation of capital within emerging technology hubs from this structural perspective, underscoring the vital role that shared institutional pathways play in mitigating market frictions.

II. Related Work

The role of venture capital within innovation ecosystems is fundamentally rooted in structural sociology and the dynamics of social capital. Ferrary and Granovetter [5] established that venture capital firms function as pivotal nodes in complex innovation networks, bridging "structural holes" between diverse institutional actors such as universities, corporations, and startups. This perspective shifts the view of VCs from mere financial intermediaries to critical orchestrators whose success depends on their embeddedness within high-trust social structures to mitigate the inherent uncertainties of early-stage investment.

Building upon these sociological foundations, empirical research has traditionally focused on syndication patterns and their impact on performance. Zheng [6] utilized social network analysis to demonstrate that a firm's centrality within co-investment networks is a strong predictor of its access to high-quality deal flow. This configuration of networks remains essential as ecosystems mature and expand globally. For instance, Aleenajitpong and Leemakdej [7] highlighted how cohesive subgroups in Southeast Asian VC networks serve as vital coping mechanisms against the information asymmetries typical of developing institutional environments. Similarly, Stohrer [8] emphasized how structural fragmentation and network strategies directly influence the performance of VC firms across the heterogeneous European landscape.

With the advancement of network science, the literature has increasingly moved toward multidimensional and predictive modeling. Esposito et al. [9] demonstrated that combining traditional network metrics with functional data allows for more robust forecasting of a startup's ability to attract continuous investment. Zhang et al. [10] further extended this by applying complex network methodologies to time-series data, revealing the dynamic evolution of attention and connectivity within these ecosystems.

Despite these advancements, a significant gap remains in the literature regarding the micro-level topology *between* individual investment managers and the founders they support. While macro-level ecosystem statistics are well-documented, the specific institutional pathways—such as shared education or prior professional overlaps—that form the "micro-foundations" of trust in multilayer contexts are under-explored. This study addresses this gap by applying advanced community detection algorithms, including the Louvain multi-level optimization [15], label propagation [17], and modularity-maximizing mergers [16], to a synthesized multilayer network. By integrating layers of education, experience, and volunteering, this study extends the literature beyond single-layer financial ties [18] and delineates a more granular, multidimensional topology of the social fabric that precedes formal investment relationships.

III. Data

A. Data Sources

The dataset for this study was constructed from two primary sources:

1) **The Portfolio Website of 212:** The official portfolio list [2] and team directory [3] of 212 provided the seed data for 54 portfolio companies and their associated investment managers. The data on the employees of 212 could be precisely constrained to the particular people who took part as investment managers responsible for the investments in portfolio companies because the firm had provided the names of investment managers in the associated webpages for the companies within their portfolio. Collectively, this data serves as the "ground truth" for the financial layer of the multilayer cofounder-investment manager network, and the fundamental source for the LinkedIn links of 212, its portfolio companies, the cofounders of these companies, and the investment managers who are associated with these companies.
2) **LinkedIn:** Public LinkedIn profiles of 49 cofounders and 10 investment managers, the links for which were collected from the associated websites of 212 and the employee search performed on the LinkedIn page of 212 [4], were systematically scraped to retrieve the educational, professional, and volunteering backgrounds of the 59 people in total.

The webscraping process for constructing the dataset was performed using a custom-built infrastructure leveraging the Playwright browser automation library and BeautifulSoup for HTML parsing. In order to ensure data privacy and adherence to ethical scraping practices, the webscraping code for the study was written to focus strictly on structural variables like institutional names, obtained degrees, and employment dates, rather than unstructured personal text content, such as the "About" sections, posts, or the recommendations by LinkedIn. In order to respect rate limits and prevent blockage during the process, the scraper code employed instantiation and destruction of concrete browser instances, as opposed to the practice of headless scraping, randomized delays between the webpage transitions and session state management, so that the access to the LinkedIn pages could be sustained by keeping the login cookies associated with LinkedIn.

B. Construction of the Dataset

The raw data extraction and network generation pipeline was organized into a multi-stage process, reflecting the transition from unstructured web content to mathematically rigorous network models. The artifacts of this process are structured into the following specific data categories:

- **Raw HTML Files (`html/`):** The collected, unstructured LinkedIn profile pages for both cofounders and 212 investment managers, securely cached as the foundational reference data.
  - **Note:** This directory is not shared with public as it contains the source code for the raw LinkedIn pages.
- **Structured Tables (`csv/`):** The parsed HTML data formatted into tabular structures mapping relational attributes. Key files include *212_cofounders_linkedin_main.csv* and *212_employee_linkedin_main.csv* for basic profiles, alongside distinct relational files tracking *education*, *experience*, and *volunteering-experiences* separately for both startup founders and investment managers. The vital portfolio assignments bridge was compiled in *212_cofounders_with_investment_managers.csv*.
- **Relational Objects (`json/`):** Aggregated mapping dictionaries consolidating the institutional affiliations and

roles of cofounders and investment managers, in a separated manner in accordance with the position of the people, to prepare the data for structural processing.

The final consolidated dataset encompasses detailed profiles for 59 unique individuals linked through shared portfolio companies (cofounded companies for cofounders; managed investments for investment managers) and common institutional backgrounds (education, work experience, volunteering). Prior to any network generation, individual names and institutional titles were tried to be normalized using string-matching algorithms to resolve minor spelling disparities and ensure reliable mappings across different records.

IV. Methodology

A. Construction of the Networks

The core analytical output of this study is a suite of exactly ten graph projections derived from the institutional and financial data, culminating in a synthesized multilayer network $M = (G, L)$. In this context, G is defined as the set of nodes representing cofounders ($n = 49$), investment managers ($n = 10$), and the formal entities (the VC firm 212 and its portfolio companies). Using the NetworkX library in Python, the data were systematically exported to the `data/gephi/` directory to construct the following configurations:

1) **The Projected Network of Cofounders and Investment Managers (`cofounders_employee.gexf`):** The primary analytical artifact of the study. This unimodal projection flattens all formal investment ties and informal institutional links into a single graph. It serves as the definitive basis for community detection, centrality rankings, and disassortativity analysis.
2) **Financial Investment Networks (`vc_company.gexf`, `company_cofounder.gexf`, `company_investment_manager.gexf`):** Three structural projections capturing the ground-truth business relationships. These establish the bipartite connective spine of the ecosystem, linking the venture capital firm (212) to portfolio companies, and subsequently bridging founders and investment managers assigned to the same ventures.
3) **Institutional Trust Networks (`{cofounders, employee}_{education, experience, volunteering-experiences}.gexf`):** Six distinct unimodal thematic network layers representing shared personal histories, isolated natively by the roles of professionals:
   - **Education:** Links between individuals sharing an alma mater, revealing long-term trust formation built during formative years.
   - **Experience:** Ties based on shared professional experience at prior corporate organizations.
   - **Volunteering:** Connections formed through shared civic or philanthropic initiatives.

In order to perform a cohesive structural analysis across the entire ecosystem, the multilayer dataset was first represented as a set of distinct relational graphs corresponding to financial ties, education, professional experience, and volunteering. These layers captured not only heterogeneous, but also complementary dimensions of connectivity among cofounders and investment managers.

For the main structural analysis, the layers of the networks were integrated into a single *collapsed weighted projection*. In this projection, the node set was preserved, while edges were defined based on the union of all relational layers: a link existed between two individuals if at least one type of relationship was present. Edge weights were assigned as the sum of binary indicators across layers, corresponding to the number of distinct relational dimensions (financial, education, experience, volunteering) shared by a given pair of nodes. This aggregation preserves multilayer information in a compact form while enabling the application of standard single-layer community detection and global network statistics. Importantly, the resulting edge weights encode the intensity of multimodal overlap, ensuring that pairs connected through multiple institutional channels contribute more strongly to structural cohesion and centrality measures.

B. Standard Analysis Pipeline

The graph structures were programmatically analyzed using the NetworkX library in Python. Standard network metrics were computed based on the theoretical framework established by Newman [11] to characterize the topology. Specifically, network sizes, density, average degree, diameter, and assortativity coefficients were measured for each isolated layer and the aggregated projection to quantitatively evaluate the core-periphery and hub-and-spoke organization. Centrality metrics, including degree, betweenness, and eigenvector centrality, were processed to identify structural brokerage roles following the conceptual clarifications established by Freeman [12]. Topological distributions, specifically the degree density and modularity scores, were computationally rendered using `Matplotlib` to ensure precise analysis. For spatial representation, models were exported as GEXF files and visualized using the ForceAtlas2 [13], optimized for handling continuous graph layouts and improving community visual modularity, and Yifan Hu [14] algorithms. The resulting layout configurations are systematically archived in the `gephi_viz/` artifacts.

C. Community Detection

To investigate the micro-foundations of the venture capital ecosystem, this study implements a community detection pipeline designed to isolate dense relational clusters within the aggregated cofounder-investment manager network. This analytical approach identifies groups with elevated internal connectivity, effectively mapping the institutional pathways and trust-based signals—such as shared alma maters or professional histories—emphasized in the structural sociology of venture capital [5]. By consolidating these diverse relational

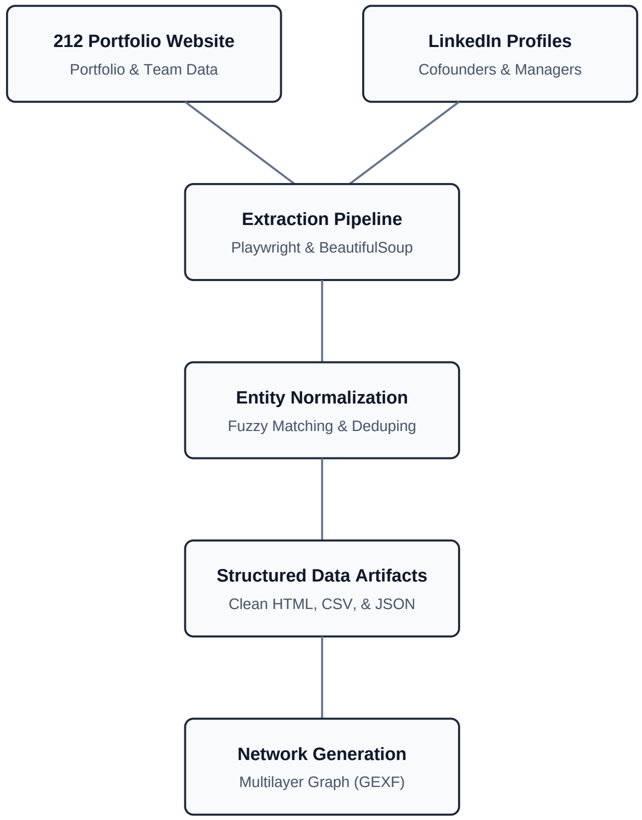
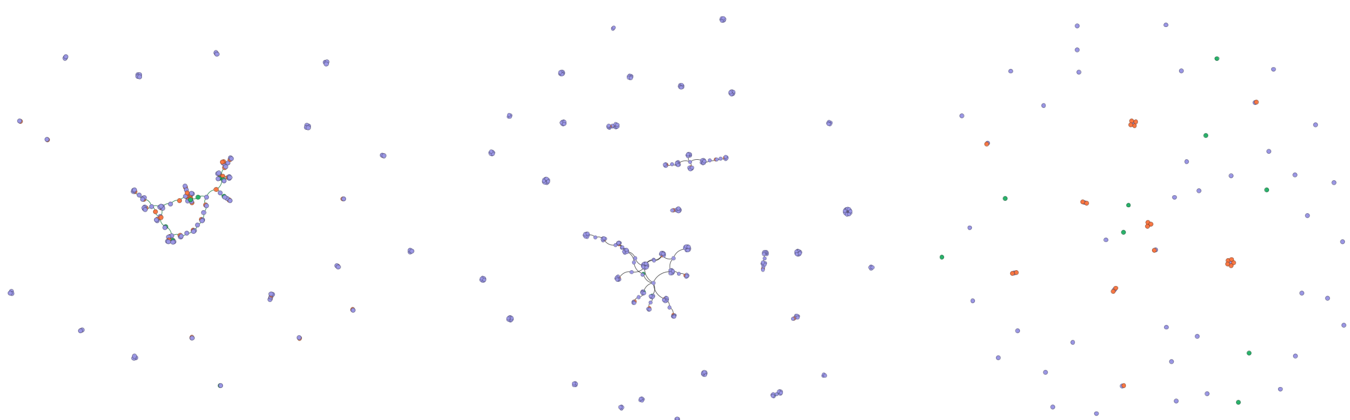


Fig. 1: Architectural overview of the data engineering and network synthesis workflow, starting from ground-truth VC portfolio datasets and LinkedIn profile extraction via Playwright, through automated entity normalization and institutional mapping, culminating in the generation of multilayer GEXF graph projections for ecosystem analysis.



(a) Education Layer (b) Experience Layer (c) Volunteering Layer

Fig. 2: The institutional layers within the dataset, comparing the topological dispersion of trust networks spatialized via the ForceAtlas2 layout. The education layer (a) exhibits denser clustering (alma mater ties) compared to the more sparse and fragmented volunteering layer (c).

modalities into a unified multilayer framework, a robust basis is provided for analyzing the modular organization and stability of the portfolio network.

In order to ensure the stability and robustness of the resulting partitions, four distinct algorithms were utilized:

- **Louvain Algorithm**: A multi-level modularity optimization algorithm effective for hierarchical community structures in large networks [15]. It greedily optimizes the modularity parameter Q.
- **Greedy Modularity**: An iterative merging approach that maximizes modularity gain at each step, providing a deterministic baseline for smaller graphs [16].
- **Label Propagation**: A fast heuristic that assigns labels based on neighbor frequency, identifying well-defined clusters without a predefined community count [17].
- **Girvan-Newman Algorithm**: A divisive method based on the removal of edges with the highest betweenness centrality, useful for identifying the structural bridges connecting communities [18].

These algorithms were applied to the aggregated network using the NetworkX library in Python. Modularity Q was used as the primary metric for evaluating partition quality:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

where A_{ij} is the weight of the edge between i and j, k_i is the degree of node i, c_i denotes the community of node i, m is the sum of all edge weights, and the Kronecker delta function $\delta(c_i, c_j)$ evaluates to 1 if $c_i = c_j$ and 0 otherwise.

In order to be able to guarantee the reproducibility of the partition results, all stochastic algorithms were executed using a standardized initialization state (`seed` $= 42$), while edge weights—reflecting the aggregate intensity of multilayer connectivity—were fully integrated into the modularity calculations. Beyond pure topological extraction, the underlying socio-technical drivers of these clusters were investigated through an attribute enrichment process. By cross-referencing the modal institutional affiliations (e.g., specific alma maters or historical professional roles) within each detected community, the partitions were mapped to real-world institutional pathways. This mixed-methods approach facilitates a deeper understanding of whether communities are driven by financial syndication or shared social capital. For visual validation, the modular structures were exported as GEXF artifacts and spatialized using the ForceAtlas2 algorithm. This layout was specifically selected for its ability to convert the density of intra-community edges into spatial proximity, providing a qualitative confirmational lens for the modularity metrics computed programmatically.

V. Results

The empirical analysis of the synthesized ecosystem reveals a highly-structured and non-random topological organization. Table I summarizes the key network parameters across different structural projections. Unless stated otherwise, all centrality and community measures on the aggregated network use the additive edge-weight scheme introduced in Section IV-A, where each of the four relational layers (financial, education, experience, volunteering) contributes equally with a binary weight of 1. The aggregated cofounder-investment manager network, which serves as the primary analytical backbone, exhibits a sparse overall density (0.054) but maintains a relatively high average degree (3.15), indicating that connectivity is concentrated within specific high-volume channels. The scale differential between institutional layers and the bipartite financial core is further visualized in Fig. 3, highlighting the expansion of the professional experience layer relative to the volunteering and educational footprints.

TABLE I: Comparative Network Topology across Layers

Layer	Nodes	Edges	Dens.	Avg. D.	Assort.
Cof.-Ind. (Agg.)	59	93	0.054	3.15	-0.718
Cof. Education	143	123	0.012	1.72	-0.282
Cof. Experience	386	358	0.005	1.85	-0.597
Cof. Volunt.	70	22	0.009	0.63	-0.489
VC-Company	55	54	0.036	1.96	-1.000
Comp.-Cof.	95	50	0.011	1.05	-0.111
Comp.-Inv.	55	47	0.032	1.71	-0.611

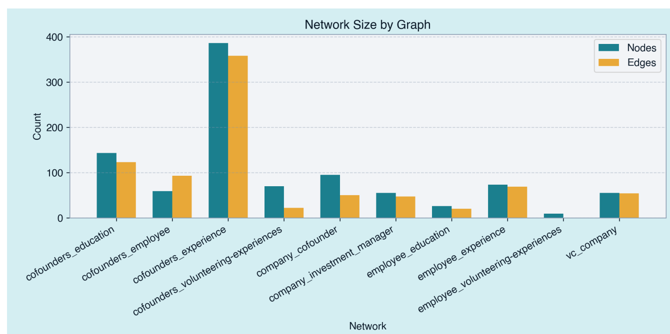


Fig. 3: Comparative distribution of node and edge volumes across the separated unimodal and bipartite topology configurations of the ecosystem, demonstrating the varying scale of social capital across institutional layers.

A. *Centrality and Brokerage*

The degree centrality rankings are dominated by the investment managers of 212, verifying their role as the primary structural gatekeepers of the ecosystem. Caglar Urcan (degree 16, betweenness 0.270), Selma Bahcivanoglu (degree 13, betweenness 0.215), and Ali Karabey (degree 13, betweenness 0.211) exhibit the highest combined centrality scores, shown in Fig. 6, effectively acting as brokers who bridge otherwise isolated cohorts of founders.

This configuration is quantitatively reflected by a strongly negative assortativity coefficient ($r = -0.718$), which is expected given the bipartite-like construction of the network — investment managers are structurally required to connect to many low-degree founders, producing degree heterogeneity

Fig. 4: Force-directed layout (ForceAtlas2) of the primary cofounder-investment manager network, exported from the Gephi workspace. The central positioning of investment managers visually confirms their role as the ecosystem's topological core, facilitating a distinct hub-and-spoke organization.

by design. While it indicates a strongly centralized connectivity pattern in the aggregated projection, this disassortativity should not be read as a purely emergent sociological signal; it is largely a consequence of how the graph was constructed. Nevertheless, the partnership structure of 212 functions as a centralized routing layer where information flow is filtered through hubs, mitigating the noise of unstructured connectivity but potentially creating single points of failure at the broker level. The relative distribution and metric comparison of these centrality scores, indicating the heavy-tailed nature of brokerage in the 212 network, are illustrated in Fig. 5.

B. Community Structure

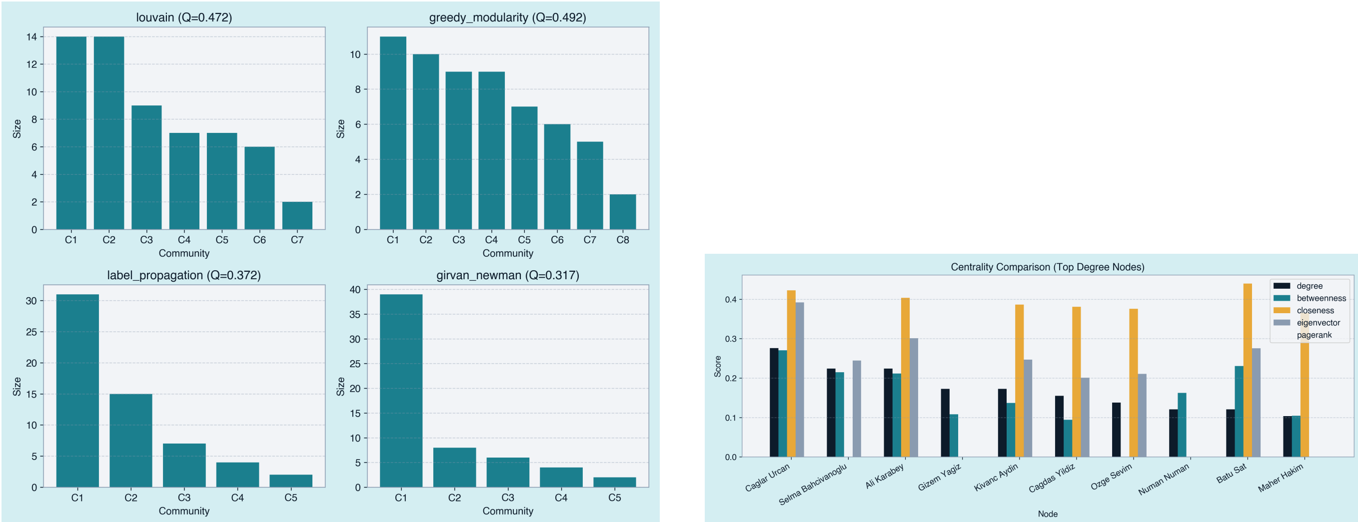
Using the Louvain multi-level optimization algorithm, seven communities are extracted with an empirical modularity score of $Q = 0.4721$, which represent the full partition of the 59-node network under the chosen resolution. This indicates the presence of a non-trivial modular organization under the chosen projection and weighting scheme. However, given the relatively small network size ($N = 59$) and the structurally induced connectivity patterns introduced by the bipartite-like linkage between investment managers and founders, modularity values should be interpreted as descriptive rather than definitive evidence of strong emergent community formation. A comparative assessment of algorithm performance (Table II) demonstrates that while alternative heuristic routines vary in their absolute community counts, the underlying localized partitions remain structurally consistent.

TABLE II: Comparative Community Detection Performance on the Collapsed Single-Layer Projection ($N = 59, M = 93$)

Algorithm	Key Parameters	Communities (k)	Modularity (Q)
Louvain (Empirical)	Weight: `weight`, Seed: 42	7	0.4721
Louvain (Dynamic Null)	1,000 Rewirings (Mean)	—	0.4565
Greedy Modularity	Weight: `weight`	8	0.4920
Label Propagation	Weight: `weight`	5	0.3722
Girvan-Newman	Split Level: 3	5	0.3167

Community detection was performed on the aggregated multilayer graph using the additive edge-weight formulation described in Section IV-A, where all four relational layers (financial, education, experience, volunteering) contribute equally to edge strength. All seven detected communities are characterised below based on their membership composition, brokerage concentration, and institutional overlap signals derived directly from the data.

- **Community 1 — The Deep-Tech Cluster (Size 7)**: Centred around Caglar Urcan, this cluster groups founders of deep-tech hardware and materials ventures including *Haelixa, One Five, Aepnus,* and *Aibuild*. Two of the seven members share a Boğaziçi University background, providing an educational anchor alongside shared investment ties.
- **Community 2 — The SaaS & Travel Core (Size 9)**: Managed by Selma Bahcivanoglu, this community encompasses founders from *Hotelrunner, Juphy, Phitech,* and *Agrotics*, among others. Two members share a Boğaziçi University background and one a Marmara University affiliation, situating the cluster within an overlapping local alumni network tied together by professional experience in the SaaS and travel-tech verticals.
- **Community 3 — The Cross-Border Scale-Up Group (Size 6)**: Centred around Kivanc Aydin, this cluster aggregates founders from *Freya* and *Qrambo* alongside the broker. A shared LSE educational background (1 member) and overlapping professional experience at Bolt Insight distinguish this community as a cross-border-oriented sub-group, reflecting a pathway from international education to Turkey-based scaling ventures.
- **Community 4 — The Investment-Anchored Mixed Sector Group (Size 7)**: Managed by Numan Numan, this cluster connects founders of *Chooch, Omma, Socradar, Trio Mobil, Artboard Studio,* and *Flow48*. The community is held together purely through shared portfolio investment ties, with no dominant shared educational institution, making it the most structurally investment-driven partition in the network.
- **Community 5 — The International Expansion Hub (Size 14)**: The largest community alongside Community 7, co-brokered by Ali Karabey and Gizem Yagiz, encompasses founders of 8 portfolio companies including *Marti, Iyzico, Insider One,* and *Avatao*. Three members share a METU background and two share LSE ties, while



(a) Community Size Distributions

(b) Centrality Score Comparison

Fig. 5: Computational topological analysis: (a) distribution of community sizes across the different algorithms highlighting partition stability; (b) relative centrality score distributions by metric, identifying the dominant structural brokerage of the central hubs.

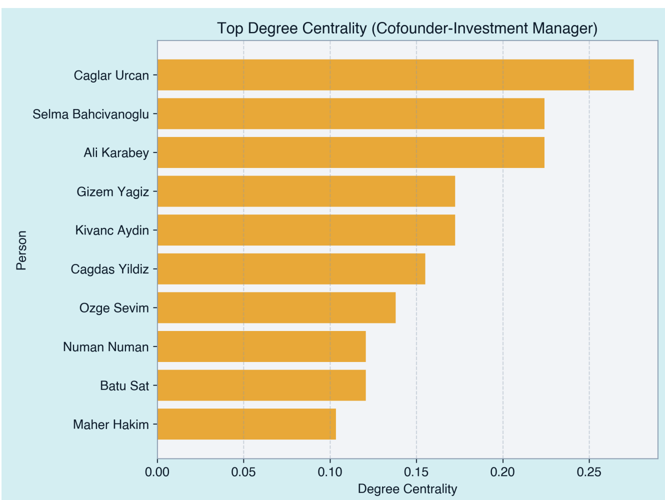


Fig. 6: Top degree centrality rankings in the multilayer network, highlighting the long tail created by a small segment of highly-connected investment managers acting as structural bridges for the 54 portfolio companies.

a Deutsche Bank professional overlap provides an additional institutional anchor. The dual-broker structure and international educational footprint make this the primary cross-border and growth-stage sub-network within the portfolio.

- **Community 6 — Isolated Dyad (Size 2)**: This minimal community consists solely of Mehmet Uygun and Cem Okay, connected exclusively through the *Getmobil* investment relationship, with no shared institutional or professional background. It represents a peripheral, weakly embedded node pair at the boundary of the ecosystem's core.

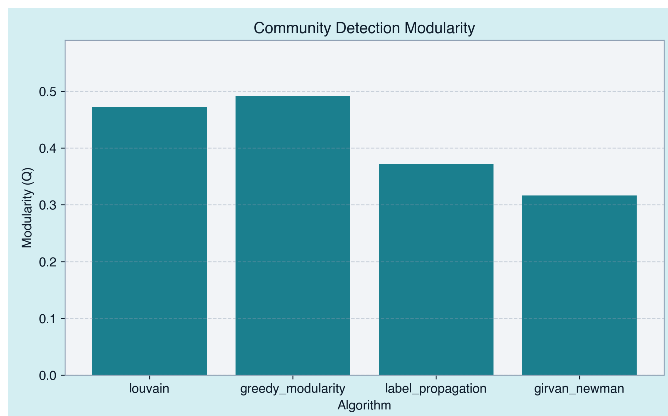


Fig. 7: Comparative modularity metrics (Q) achieved by disparate partition optimization strategies, highlighting the stability of the detected ecosystem sub-clusters.

- **Community 7 — The ITU-Anchored Technology Cluster (Size 14)**: Co-brokered by Cagdas Yildiz, Maher Hakim, and Ozge Sevim, this community encompasses founders of *123Formbuilder, B2Metric, Evercopy, Meddy, Wellbees,* and *Philosopherking*. It is the most strongly educationally anchored community in the partition: 10 members share an Istanbul Technical University (ITU) background, making it the single densest alma-mater cluster in the network. Carnegie Mellon University ties (1 educational, 1 professional) and Wellbees as a shared employer provide additional cross-layer cohesion.

C. Methodological Validation via Dynamic Null Model

To validate whether the partitions discovered by the Louvain algorithm reflect genuine socio-economic clustering or are merely structural artifacts of the network's degree distribution, a dynamic null model benchmark is implemented. This step directly addresses topological concerns regarding the hub-and-spoke layout inherently generated by major investment managers. Rather than evaluating the observed community boundaries on a static randomized graph, a method that artificially inflates significance metrics, the algorithmic optimization capability itself is tested.

A total of $n = 1,000$ randomized counterpart graphs are generated utilizing a double-edge-swap framework that preserves the exact degree sequence of the collapsed single-layer network projection ($N = 59, M = 93$). For each randomized iteration, the Louvain community detection algorithm is executed fresh with an independent seed parameter. This creates an optimized null distribution that maps the maximum modularity potential of a random network possessing identical degree constraints.

Under this dynamic configuration model, the randomized variations achieve a high mean optimized modularity of $\mu_{\text{null}} = 0.4565$ ($\sigma_{\text{null}} = 0.0140$, ranging from a minimum of 0.4162 to a maximum of 0.5054). This yields a tight modularity gap of:

$$\Delta Q = Q - \mu_{\text{null}} = 0.0156$$

and a corresponding z-score of:

$$z = \frac{\Delta Q}{\sigma_{\text{null}}} = 1.1143$$

This evaluation offers a critical structural insight. The elevated baseline mean ($\mu_{\text{null}} = 0.4565$) confirms that a significant portion of the network's macro-level modularity is driven by its raw topological boundaries—specifically, the degree heterogeneity of the investment hubs. The empirical network does outpace the re-optimised random baseline ($\Delta Q > 0$), but the z-score of 1.11 falls short of the conventional threshold of $z \geq 2$ for statistical significance. This result should be interpreted with care: it does not falsify the community structure, but it does mean that the detected partition cannot be distinguished from degree-sequence artefacts with high confidence at this network size. The honest reading is that the macrostructure is strongly constrained by the portfolio's hub-and-spoke architecture, while the specific edge configurations within real-world modules—driven by genuine relational overlaps such as shared alma maters and professional histories—provide a modest but non-zero coordination signal beyond what random rewiring produces.

D. Comparative Algorithmic Analysis

To ensure that the structural conclusions are robust across differing optimization mechanics, the collapsed single-layer graph is evaluated against alternative community detection routines. As detailed in Table II, the greedy modularity maximization framework yields the highest absolute partition score ($Q = 0.4920$) across 8 distinct clusters. This algorithm behaves aggressively, isolating highly dense local cliques formed by overlapping historical affiliations.

Conversely, algorithms that prioritize localized neighbor consensus over global modularity targets yield more aggregated structures. Label propagation stabilizes into 5 core communities ($Q = 0.3722$), dominated by a single large macro-component of 31 closely linked individuals. This macro-cluster effectively highlights the deep, cross-layer systemic integration within the broader ecosystem, where educational ties and shared professional backgrounds bridge across distinct investment portfolios.

Finally, the hierarchical Girvan-Newman edge-betweenness split reaches an optimal modularity plateau at level 3 ($k = 5, Q = 0.3167$). It successfully isolates marginal, lower-density structures—such as the explicit two-node dyad in Cluster 5 consisting of a single founder and an investment manager—before fracturing the highly resilient core components of the network. The recurring alignment of specific institutional blocks across these diverse algorithmic assumptions indicates that the broad community boundaries are a consistent structural feature of the multi-layer dataset, even if their statistical separation from degree-sequence effects is modest at this network scale.

VI. Discussion

The structural analysis of the 212 venture capital ecosystem suggests a non-random and highly centralized topology that differs from what is typically observed in organically formed social networks.

A. Brokerage and Managed Topology

The negative assortativity coefficient ($r = -0.718$) reflects a pronounced hub-and-spoke structure in the aggregated projection, where a small number of high-degree investment managers connect otherwise sparsely linked groups of founders. This pattern is substantially induced by construction: investment managers are, by definition, bridging nodes across multiple portfolio companies, making degree heterogeneity a structural expectation rather than a discovered property. Consequently, the observed disassortativity should be interpreted as a confirmation that the network was built correctly, not as evidence of an emergent organizational logic beyond what the bipartite architecture guarantees. The more analytically meaningful question—addressed in the null model benchmark—is whether edge placement within that degree-constrained skeleton is non-random. While typical human social networks often exhibit positive assortative mixing driven by homophily, professional venture capital networks are structurally oriented toward brokerage and information mediation. In this context, the partnership structure of 212 operates as a centralized routing layer in which investment managers act as intermediaries between founder communities, facilitating connectivity while also concentrating brokerage roles and introducing potential central points of dependency in information flow.

B. Institutional Trust as a Connectivity Catalyst

The modularity analysis confirms that educational overlap serves as a primary structural "glue," bridging the gap between informal social capital and formal financial ties. The recurrence of local institutions, specifically Boğaziçi University, Middle East Technical University (METU), and Istanbul Technical University (ITU), within the detected Louvain modules suggests that regional trust networks remain phenomenologically vital even for firms with global horizons. Notably, Community 7 is the most educationally concentrated partition in the network, with 10 of its 14 members sharing an ITU background, underscoring the depth of that alumni pipeline. These institutional pathways operate as proxy due-diligence mechanisms; shared backgrounds allow investors to bypass the high frictions of early-stage uncertainty through implicit reference checks. Conversely, the "bridge" effect driven by the London School of Economics (LSE)—which appears in Community 5 (two members) and Community 3 (one member)—provides the structural openings necessary for cross-border expansion, suggesting a dual-tier strategy of local trust and global reach.

C. Multilayer Synergies and Information Asymmetry

A key revelation of approaching the portfolio as a multilayer network is the synergy between professional experience and financial ties. While the education layer provides the initial trust-based entry points, the experience layer (the largest by edge volume, as shown in Table I) provides the functional density required for sustained collaboration. The community evidence is consistent with the view that cofounders who share both an alma mater and prior professional contexts tend to reside in the same detected community—for instance, Community 5 members share both METU ties and Deutsche Bank experience. This pattern suggests that investment managers at 212 are not merely selecting companies, but navigating pre-existing clusters of professional social capital to mitigate information asymmetries, though a causal claim would require longitudinal data beyond the scope of this study.

D. Implications for the Ecosystem

For **entrepreneurs**, the robust modularity observed in the cofounder aggregation suggests that strategic alignment with specific institutional clusters is not merely a social advantage but a structural requirement for visibility. For **venture practitioners**, these results illuminate the latent biases inherent in portfolio construction; an over-reliance on established institutional pipelines may lead to a homogenization of human capital, potentially overlooking high-potential founders from outside traditional "alma mater hubs." Understanding these topological signatures allows for a more deliberate approach to ecosystem diversification.

VII. Limitations and Future Work

A primary limitation of this study is the relatively small scale of the dataset, which is constrained to a single VC firm's portfolio. With only 59 nodes and 93 edges, the resulting network is sparse, and both modularity and assortativity measures are sensitive to small structural perturbations. In particular, the presence of high-degree investment managers structurally biases global statistics such as disassortativity and community separability, making it difficult to fully disentangle empirical effects from construction effects. Accordingly, the reported modularity ($Q = 0.472$) should be interpreted as an indication of structured connectivity under this specific dataset rather than strong evidence of universally stable community structure in venture capital ecosystems. In prospective future research, it would be valuable to expand this analysis to multiple VC firms, which would help determine if these structural patterns are common across different investment strategies or unique to this specific portfolio.

Another significant limitation of this study is its reliance on public LinkedIn data, which is inherently self-reported and may suffer from survivor bias, incomplete coverage, and temporal inconsistency, as profiles reflect self-reported and periodically updated information. Such inaccuracies can lead to the omission of critical ties or the inclusion of obsolete affiliations, with concrete downstream effects on the detected community structure. For example, if a cofounder omits a prior degree from their profile, the corresponding education-layer edge is silently absent, which can artificially isolate two nodes that are genuinely connected—raising the inter-community gap and inflating modularity scores. Conversely, an outdated professional affiliation that no longer reflects actual ties introduces spurious edges that may incorrectly merge two otherwise distinct communities. Community 4, which is held together entirely by shared investment ties with no detectable educational or professional overlap, is a case where such data gaps could be particularly misleading: the absence of institutional signals may reflect genuine lack of overlap, or simply incomplete LinkedIn profiles. Furthermore, the bipartite nature of the investment layer means that ties between cofounders of different companies are only visible through their shared institutional backgrounds, potentially underestimating the density of the actual, unstructured social network, such as informal mentorships. To address these gaps, future work could combine scraped data with verified corporate registries or direct surveys to obtain a more complete picture of these professional ties.

Beyond data availability, the study faced significant data formatting and entity resolution challenges. The raw scraped data contained numerous naming irregularities for portfolio companies (e.g., *SOCRadar* vs. *Socradar*) and inconsistent institutional titles, which required extensive string-matching and manual normalization to avoid creating duplicate nodes for the same entity. In addition to that, some profiles were entirely missing or inaccessible due to LinkedIn's privacy settings or account deletions, leading to a slight under-representation of the total possible network size. To improve this, future iterations could employ a more systematic approach to entity resolution, such as using a standardized institutional gazetteer or a more rigorous set of normalization rules to handle common naming variations.

Additionally, specific data integrity challenges were encountered during the scraping of the ground-truth portfolio datasets. These include missing profiles for one investment manager, for whom not any link could be obtained from either the website of 212 or LinkedIn, naming irregularities for portfolio companies across pages, for example, *SOCRadar* and *Socradar*, both of which indicate the same company, and an occasion of a link inconsistency in which the portfolio entry of 212 for *iyzico* provided information for *Insider One* (formerly known as *Insider*) instead. While the current study prioritized a robust automated pipeline for data generation, these anomalies highlight the need for enhanced entity resolution and manual verification in future iterations. A practical next step would be to introduce a human-in-the-loop verification step, such a simple interface where a researcher can approve or merge suspected duplicate entities (like *iyzico* and *Insider*), so as to catch and fix these kinds of inconsistencies before the network is actually generated.

Another aspect that should be incorporated by prospective future work on this study is temporal data so as to investigate whether networks grow denser or sparser dynamically around funding rounds. In addition, incorporating co-investment data with entirely separate competitor VCs would illuminate the bridge connectivity across the wider Turkish startup ecosystem at large. Expanding the dataset to include multiple VC firms operating within Turkey would enable a comparative structural analysis of portfolio strategies and institutional biases across the industry.

VIII. Conclusion

This study has demonstrated the utility of multilayer network analysis in deciphering the complex social fabric governing venture capital portfolios. By synthesizing institutional and financial data from the 212 ecosystem, a highly centralized, disassortative topology ($r = -0.718$) was identified, where investment managers function as critical structural bridges between isolated founder communities. It must be noted, however, that this disassortativity is substantially a consequence of the bipartite investment-manager–founder design rather than a purely emergent finding; the null model benchmark ($z = 1.11$) likewise indicates that the observed modularity cannot be fully disentangled from degree-sequence constraints at this network scale. Within these interpretive bounds, the findings reveal that communities are not random aggregations but are meaningfully reinforced by pre-existing institutional pathways, with local alma mater networks—specifically Boğaziçi, METU, and ITU—serving as primary catalysts for trust and connectivity within the Louvain-detected communities. Conversely, international institutional bridges like LSE provide structural openings for cross-border scalability, as evidenced in Communities 3 and 5.

Within the specific context of the 212 portfolio, the results suggest that professional social capital is closely associated with founder visibility and connectivity. Although the dataset is limited to a single venture capital firm and a relatively small sample of actors, the reproducible methodology provides a framework for examining how institutional relationships shape startup ecosystems. More broadly, the findings illustrate how shared educational and professional histories may contribute to trust formation and the emergence of formal investment relationships, while motivating future comparative studies across larger and more diverse venture capital networks.

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APPENDIX A
GRAPH VISUALIZATIONS REFERENCE

In addition to the primary figures included throughout the text, a comprehensive suite of computational layouts covering all isolated thematic and bipartite structural components natively generated via ForceAtlas2 and Yifan Hu algorithms has been systematically exported. These reference variants can be found in the `gephi_viz/exported/` directory. Key architectural mappings include:

- **Financial Connectivity:** `vc_company_cofounder`, `vc_company_inv_manager`, `company_cofounder_inv_manager`.
- **Institutional Trust Pathways:** `education`, `experience`, `volunteering`.
- **Aggregated Ecosystem Layouts:** `cofounders_employee` and `all_networks_combined`.

APPENDIX B
SUPPLEMENTAL STATISTICAL TABLES

Table III provides a detailed breakdown of the top influencers within the cofounder-investment manager network, highlighting the brokerage role of 212 partners.

TABLE III: Top Brokerage and Influence Rankings (Main Network)

Node Name	Type	Degree	Betweenness
Caglar Urcan	Manager	16	0.270
Selma Bahcivanoglu	Manager	13	0.215
Ali Karabey	Manager	13	0.211
Gizem Yagiz	Manager	10	0.108
Kivanc Aydin	Manager	10	0.137
Batu Sat	Cofounder	7	0.230
Sean Yu	Cofounder	6	0.130
Numan Numan	Manager	6	0.162
Cagdas Yildiz	Manager	6	0.094
Maher Hakim	Manager	6	0.105

The following clusters (Table IV) were identified using the Louvain algorithm ($Q = 0.472$), showing the institutional focus of each sub-group within the 212 portfolio.

TABLE IV: Characterization of Detected Ecosystem Communities (Louvain, $N = 59$)

ID	Size	Primary Broker(s)	Defining Signal / Shared Background
1	7	Caglar Urcan	Deep-Tech / Boğaziçi Alumni (2/7)
2	9	Selma Bahcivanoglu	SaaS & Travel / Boğaziçi & Marmara
3	6	Kivanc Aydin	Cross-Border Scale-Up / LSE Tie & Shared Experience
4	7	Numan Numan	Mixed-Sector / Investment-Only Ties
5	14	Ali Karabey, Gizem Yagiz	International Expansion / METU & LSE Pathways
6	2	Cem Okay	Isolated Dyad (Getmobil)
7	14	Cagdas Yildiz, Maher Hakim, Ozge Sevim	ITU-Anchored Tech Cluster / CMU Experience

APPENDIX C
SUPPLEMENTAL COMPUTATIONAL TOPOLOGY

Fig. 8 showcases the absolute structural complexity of the 212 ecosystem when all institutional and financial layers are aggregated into a single multilayer projection. Fig. 9 reveals the tripartite core, while Fig. 10 illustrates the scaling properties of average degree and density across these isolated layers.

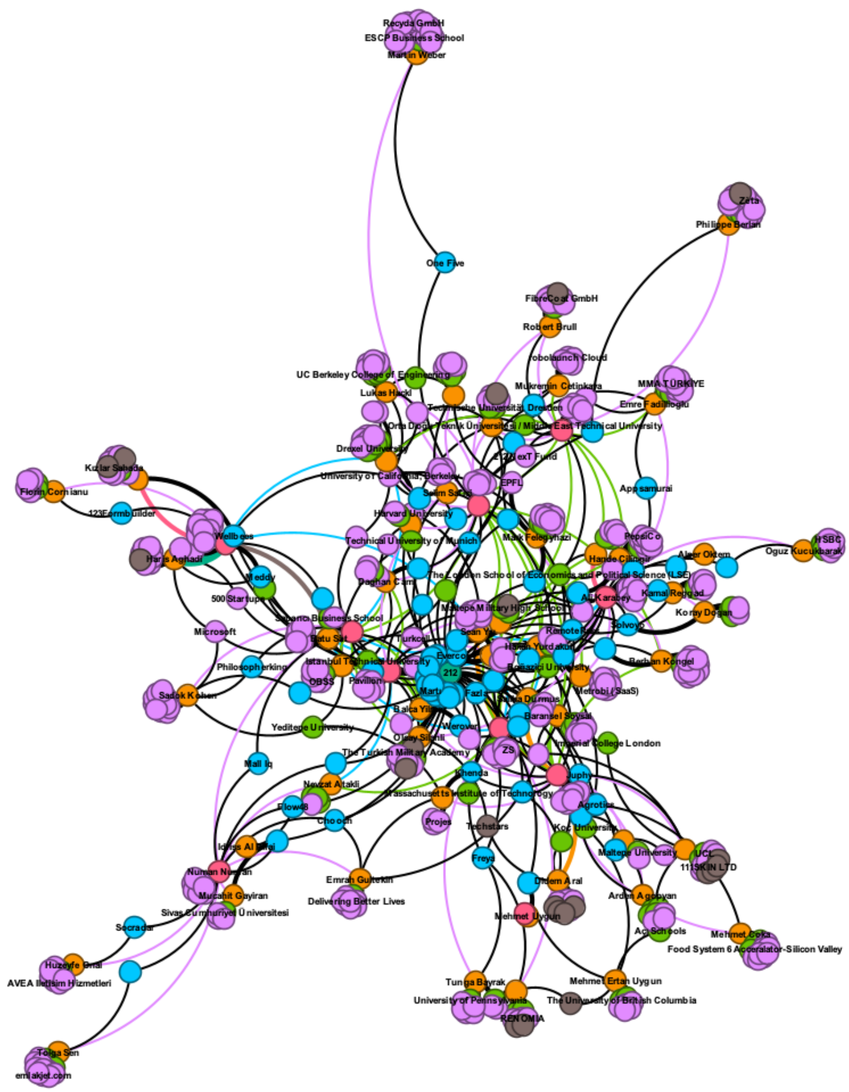

Fig. 8: Global multilayer aggregate of the 212 ecosystem, featuring all nodes (Cofounders, Companies, Investment Managers) and edges (Financial, Education, Experience, Volunteering) spatialized via ForceAtlas2.

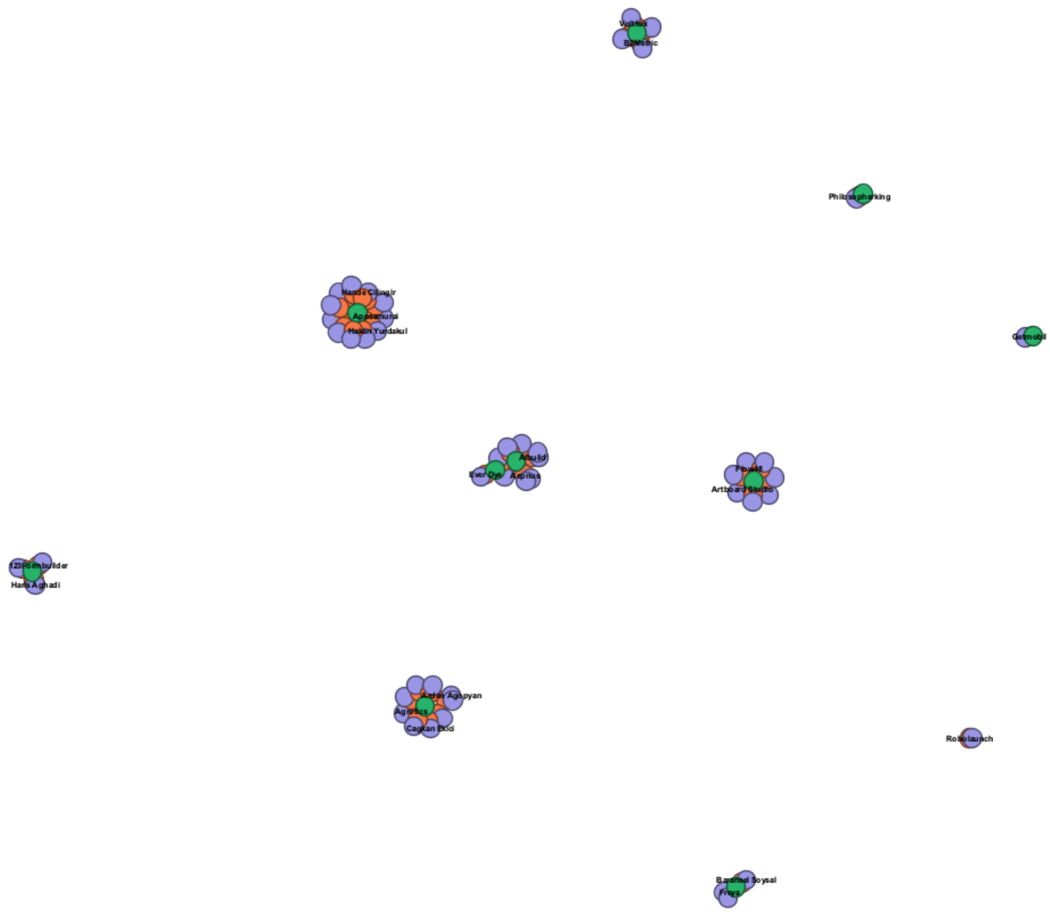

Fig. 9: The cofounder-company-investment manager tripartite core, highlighting the shared portfolio ties that form the financial backbone of the network.

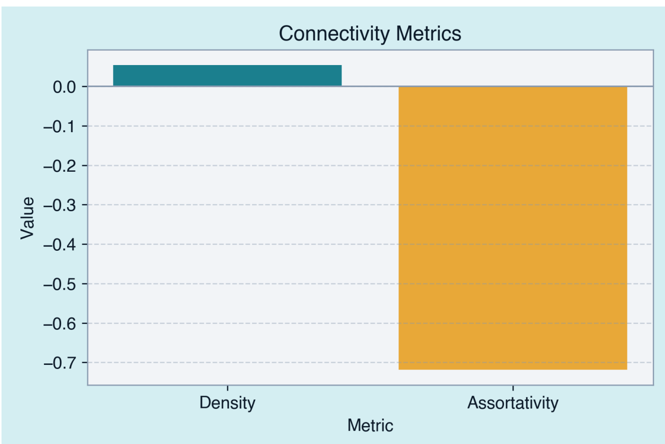

Fig. 10: Comparative analysis of average degree and network density across specific institutional layers, demonstrating the structural density of the professional experience tier relative to education and volunteering.